

HeatFlow: A Thermal-Tactile Display for Dynamic 2D Thermal Movements

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Figure 1: HeatFlow is a thermal display that creates dynamic temperature patterns by combining thermal and tactile feedback. We show three form factors: i) a $9\text{cm} \times 9\text{cm}$ forearm display, ii) a $6\text{cm} \times 6\text{cm}$ palm-sized display, and iii) a pair of two $9\text{cm} \times 6\text{cm}$ modules for the face.

Abstract

We introduce HeatFlow, a thermal display capable of generating dynamic thermal movements on a 2D surface by integrating thermal and tactile sensations. Leveraging a perception-driven approach, we combine vibration-induced thermal referral and apparent tactile motion to create the illusion of moving thermal flows. The system features a 2D array of nine tactile actuators paired with a single nearby thermal actuator. To validate our approach, we conducted three user studies. The first study determined the optimal parameters for perceivable 2D thermal flows by examining different array sizes and motion speeds under varying temperature conditions. The second study focused on optimizing our algorithm to produce smooth and continuous curved thermal flows. Comparative evaluations against two prior tactile motion techniques demonstrated that our algorithm outperforms existing methods in generating realistic thermal motion. Finally, we integrated HeatFlow into virtual

reality (VR) environments, assessing its feasibility in six interactive scenarios across three body sites—the arm, palm, and face. Our findings highlight the potential of HeatFlow to enhance immersive VR experiences by providing realistic thermal feedback across multiple body locations. Additionally, we present HeatFlow as a design tool—an application that enables users to create custom thermal flows by adjusting key parameters such as motion speed, intensity, and temperature. This work lays the foundation for more advanced thermal interfaces in interactive systems.

CCS Concepts

• **Human-centered computing** → Haptic devices; Systems and tools for interaction design; Virtual reality; • **Hardware** → Emerging technologies.

Keywords

Thermal-tactile Interface, 2D Thermal Motion, Thermal Referral

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1 Introduction

Thermal feedback is emerging as a compelling modality for enhancing immersive experiences in virtual and augmented reality. From the radiating warmth of sunlight to the sharp chill of wind, dynamic thermal sensations carry rich contextual, emotional, and environmental information. Integrated into VR, they can transform digital interactions into lifelike, emotionally resonant experiences.

Imagine a VR game where a gunshot produces a searing streak of heat across the arm, or a training simulation where a cold compress wraps around a wound. In a horror experience, a slow, creeping chill along the spine can heighten suspense. In sci-fi worlds, users might feel energy pulses flowing outward from their body, while in wellness applications, a warm touch gently gliding down the forearm can evoke comfort and calm. These experiences all rely on a critical capability: the ability to generate fluid, directional thermal motion across the skin, not just static heat at a single point.

One approach to achieving this effect is brute force: placing dense arrays of thermal actuators to physically render heat and cold across space. But this leads to bulky, power-hungry, and impractical systems. More recent work [45] has introduced perceptual illusions that combine thermal and tactile cues to simulate 1D motion along a linear path with minimal hardware. While these are promising, they remain fundamentally limited: real-life thermal experiences rarely move in straight lines. Fire swirls, wind arcs around the body, and touch often follows curved or branching paths – forms of motion that 1D thermal illusions simply cannot express.

While prior work has demonstrated the feasibility of 1D thermal illusions by combining localized heating with tactile cues to simulate motion along a straight path, extending these techniques to 2D space is far from straightforward. Simply mapping 1D algorithms onto a plane fails to account for the complex interplay of spatial diffusion, thermal latency, and multisensory integration in thermal perception. Unlike tactile stimuli, which are spatially precise and temporally sharp, thermal signals spread slowly, linger unpredictably, and lack fine spatial resolution—especially when applied through the skin. This makes it particularly challenging to evoke smooth, continuous motion across a 2D surface, where both the direction and timing of sensations must be carefully coordinated. Moreover, existing algorithms are typically optimized for fixed linear paths and do not generalize well to complex trajectories. When applied to curved or intersecting paths, they often produce fragmented or ambiguous sensations that break the illusion of motion and diminish user immersion. As a result, current approaches fall short of enabling expressive, curved, and adaptive thermal flows needed for compelling multisensory experiences.

We introduce HeatFlow, a novel thermal-tactile interface that renders perceptually continuous and expressive 2D thermal motion across the skin using minimal hardware (see Figure 1). The system consists of a 3×3 grid of vibrotactile actuators and a single thermal actuator positioned adjacent to the array. A novel control algorithm drives HeatFlow’s ability to dynamically coordinate thermal masking and apparent tactile motion, enabling fluid temperature movements to follow arbitrary 2D trajectories – including curves, turns, and variable intensities. This approach achieves a broad range of expressive thermal illusions while maintaining a compact, low-power, and wearable form factor.

To support the creation and customization of these experiences, we developed the HeatFlow Design Tool – an interactive authoring interface that allows users to sketch, sequence, and preview thermal flow patterns. Designers can define custom paths, modulate speed and intensity, and immediately evaluate the perceptual result. Additionally, user-designed flows can be directly synchronized to audio-visual content for applications like immersive storytelling, mindfulness routines, affective communication. The tool enables rapid prototyping of multisensory interactions and makes expressive thermal feedback accessible to application developers, researchers, and artists alike.

The main contributions of this paper include: (1) a thermal-tactile system that enables perceptually continuous 2D thermal motion with minimal hardware; (2) a new control algorithm that coordinates tactile and thermal cues to simulate arbitrary thermal paths across the skin; (3) an interactive design tool for authoring custom thermal experiences; and (4) three user studies that evaluate perceptual parameters, compare algorithmic performance, and demonstrate the system’s versatility across multiple body sites and immersive VR scenarios.

2 Related Works

2.1 Thermal Referral

Thermal perception is a vital component of human sensory experience, enabling us to detect and respond to temperature changes in our environment. It is mediated by heat- and cold-sensitive thermoreceptors in the skin, which sense deviations from baseline skin temperature and activate regions such as the insular cortex [7, 25, 50, 52, 64]. Compared to tactile perception, which relies on fast-responding mechanoreceptors, thermal perception is slower—typically taking 0.5 to 2 seconds—and offers lower spatial resolution due to the diffusive nature of thermal energy [8, 23]. Despite these differences, thermal and tactile stimuli frequently co-occur in everyday life. However, the precise mechanisms underlying their integration remain unclear [2, 16, 17]. Thermal perception is also influenced by visual cues [18, 60], highlighting the complexity of multisensory integration. Illusions such as thermal referral and thermal masking offer insight into how the brain combines these modalities to construct unified thermal experiences [26].

Thermal referral is a well-documented phenomenon in which the perceived location of a thermal stimulus is altered to match the position of a concurrent tactile stimulus, creating the illusion of multiple thermal sources. Originally identified by Green [16, 17], this illusion has since been examined under various conditions—including with cold stimuli [3, 21], in the absence of touch [5], and with varying thermal intensities [59]. While the underlying mechanisms remain uncertain, thermal referral has informed the design of thermal feedback systems and haptic interfaces [34, 47]. More recently, thermal masking has emerged as a related but distinct phenomenon, in which a strong tactile stimulus masks or shifts the perceived location of thermal input [56]. This enables the illusion of localized thermal sensations at specific tactile sites, even when the actual heat source is displaced. Together, these illusions reveal powerful principles of cross-modal perception that can be leveraged to create expressive, immersive thermal-tactile experiences in interactive systems.

2.2 Sensory Motion

Perceived motion is a compelling phenomenon across sensory modalities and plays a key role in creating immersive experiences. In the tactile domain, apparent tactile motion refers to the illusion of continuous movement across the skin, generated by sequential activation of stationary actuators [30, 43]. A well-known example is the Tactile Brush [22], which creates 2D motion by inducing phantom sensations—illusory points perceived between actuators through intensity modulation [1]—and utilizing stimulus onset asynchrony (SOA) with overlapping actuator activation patterns. Park et al. [38] refined this method to improve the sharpness and clarity of motion. Beyond vibrotactile arrays, researchers have explored alternative methods for rendering tactile motion, such as pneumatic inflatables [63], air cannons [49], and tapping-based systems [32]. These approaches demonstrate the versatility and expressive range of tactile motion illusions in haptic interface design.

More recently, researchers have begun to explore illusory motion in the thermal modality. Gongora et al. [15] demonstrated inter-manual thermal motion using pulses from Peltier modules on the palm, while Chen et al. [6] investigated similar effects around the face. Moesgen et al. [36] examined how varying SOA influences thermal motion perception on the arm. These techniques typically rely on multiple thermal actuators arranged spatially to simulate directional flow. More recently, Singhal et al. [45] introduced a novel approach combining thermal masking with tactile motion, successfully evoking the illusion of thermal motion using just a 1D array of vibrotactors and a single thermal actuator. This work demonstrates the feasibility of achieving expressive thermal movement with minimal hardware—a direction we aim to extend through our system by leveraging cross-modal illusions and spatiotemporal control.

2.3 Thermal User Interfaces

Interest in thermal interfaces has surged as VR and immersive haptic technologies become increasingly accessible to consumers. The design space for these devices is extensive, encompassing both contact-based systems that use conduction and non-contact methods that employ convection or radiation. Among contact-based devices, thermoelectric Peltier modules are particularly popular due to their rapid heat generation and versatility. The size and number of Peltier modules can be adjusted to control the area of thermal stimulation [10, 27, 39], and their adaptable form factor makes them ideal for integration into a variety of wearable interfaces across unique body sites and topologies.

With the growing importance of thermal feedback, a range of wearable thermal interfaces have emerged for various use cases, body sites, and haptic patterns [14, 31, 61, 62]. Liu et al. [34] introduced ThermoCaress, a sleeve using heated water and airbags to create moving sensations on the forearm. Sun et al. [48] developed an ultrathin, skin-conforming interface for multimodal feedback. Wang et al. [55] and Yi et al. [65] used thermal referral to deliver hand-based cues while minimizing hardware obstruction. Research into thermal interfaces is extensive, and the resulting insights on device design and human perception remain highly relevant today. Recent work has shifted toward novel form factors [4, 29, 40, 42], affective haptics [11, 12, 33, 51], and immersive VR integration [28, 46, 54, 58].

3 System Design: HeatFlow

To enable expressive and continuous thermal motion across the skin, we designed HeatFlow, a thermal-tactile system that creates the illusion of 2D temperature flows using minimal hardware and perceptual coordination. Unlike prior approaches that rely on either dense actuator arrays or simplified 1D illusions, HeatFlow combines a single thermal actuator with a 3×3 grid of vibrotactile actuators and a novel control algorithm that synchronizes these two modalities in both space and time.

The design of HeatFlow is grounded in the observation that thermal perception can be spatially redirected and temporally modulated through tactile stimulation. This enables us to simulate the experience of heat or cold moving along curved, continuous paths without physically heating every point along the way. The result is a system that is lightweight, scalable, and perceptually powerful, capable of rendering realistic temperature motion on a variety of body surfaces — from the forearm to the face — in support of immersive multisensory applications.

We describe the design of HeatFlow in three parts: the goals and challenges that motivated its architecture, the hardware layout, and our algorithm for generating 2D thermal flows.

3.1 Design Goals and Challenges

Our goal with HeatFlow is to enable realistic, expressive 2D thermal motion across the skin using a wearable, low-power system. While prior work has demonstrated the feasibility of simulating 1D thermal motion using tactile cues and a single heat source, these approaches cannot generalize to curved or arbitrary 2D trajectories. The core challenge lies in the inherent mismatch between tactile and thermal feedback: thermal sensations are slow to activate, diffuse over time and space, and lack sharp spatial resolution — making them difficult to control with the precision needed for 2D flow illusions.

In contrast, tactile stimuli are fast and spatially crisp. Achieving coherent 2D thermal motion thus requires not only careful coordination between modalities but also a control framework that predicts and compensates for temporal offsets due to thermal latency. Furthermore, the system must do this without dense actuator arrays, to ensure wearability and scalability.

HeatFlow is designed to meet these goals by leveraging cross-modal perception, specifically vibration-induced thermal referral and apparent tactile motion, to simulate continuous 2D thermal motion using a single thermal actuator and a 3×3 grid of vibrotactile motors. The system is driven by a novel algorithm that dynamically blends tactile and thermal cues to maintain the illusion of temperature flowing across the skin along any user-defined path.

This mechanism is grounded in the established principles of vibration-induced thermal referral [56] and apparent tactile motion [22], and is realized through the precise spatiotemporal control required to simulate continuous 2D thermal flows. By spatially redirecting thermal perception and temporally modulating tactile cues, HeatFlow orchestrates both modalities to create coherent and expressive thermal motion — overcoming the inherent delays and spatial imprecision of thermal feedback through carefully sequenced, perceptually grounded control.

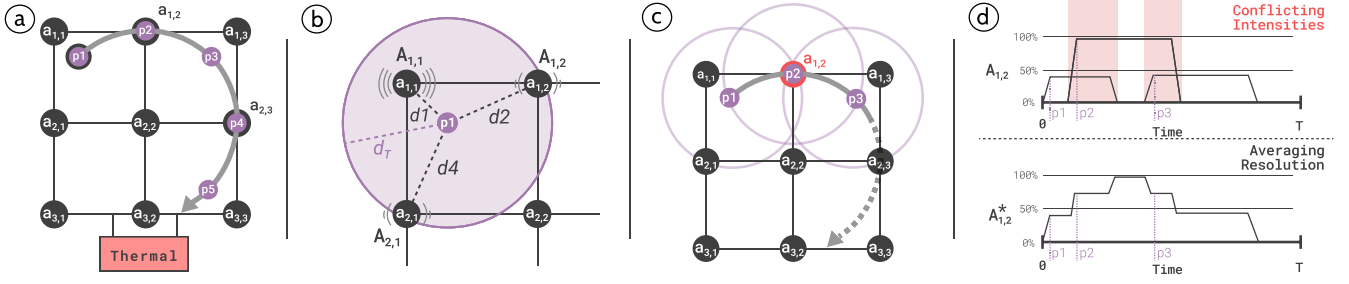


Figure 2: Concept diagrams visualizing the steps of the HeatFlow algorithm. Given a flow path, the algorithm (a) selects phantom points at equidistant intervals along the path, (b) sets target intensities for every tactile actuator within a distance threshold to create the phantom point, (c) finds actuators with conflicting intensities throughout the flow, and (d) resolves the intensity conflicts by an averaging approach.

3.2 Hardware Setup

The HeatFlow prototype is a flexible, grid-based display designed to render continuous 2D thermal motion on the skin through the integration of a single thermal actuator and a 3×3 array of vibrotactile actuators. This compact configuration is intentionally designed to balance perceptual richness, user comfort, and system simplicity — enabling dynamic multisensory feedback without the need for bulky or high-density hardware.

Grid Design and Body Adaptability. The actuator grid is 3D-printed using flexible ABS plastic, making it adaptable to various body surfaces, including the forearm, palm, and face. The lines of the grid structure maintain a thickness of 1 mm for flexibility, and the housings at each point of the grid are 5 mm thick to secure the tactile actuators with direct skin contact. While our primary setup uses a 3×3 grid, we also explored other configurations (e.g., 3×2 layouts) to assess generalizability.

To ensure reliable contact across various body parts, we used a Velcro strap system with elastic backing, allowing the grid to conform securely to curved and mobile areas such as the forearm, palm, and side of the face. The grid is inserted into a 3D-printed backing plate with built-in guide slots for the straps, which can be adjusted for different body sizes. For facial placement, the strap is routed around the head and supported with light padding to maintain comfort and stability. This setup enables consistent contact pressure across trials while remaining unobtrusive and comfortable for users.

Tactile Actuators. We employ coin-shaped eccentric rotating mass (ERM) actuators (JIEYI JYC1027), each 10 mm in diameter and 2.7 mm thick, capable of producing vibrations at 215 Hz with a peak force of 9.8 mN and voltage of 3 V. These actuators are placed at a 3 cm inter-actuator spacing, based on our empirical findings that this distance provides the best balance between spatial resolution and comfort for 2D tactile motion illusions. The equidistant spacing ensures even coverage and consistent perception across the grid surface.

Each actuator is individually controlled using pulse-width modulation (PWM), allowing fine-grained control over both vibration intensity and duration. To simulate a moving phantom point, the actuators are activated in sequence with precise temporal offsets,

producing the illusion of motion across the grid. The system dynamically computes these sequences in real time based on the user-defined path, target speed, and local geometry, ensuring the rendered motion is smooth, continuous, and directionally coherent. The motion parameters produced by the HeatFlow algorithm are hardware-agnostic, and we realized the perceived intensity values in our ERM-based implementation by mapping input voltage via a power-law relationship with measured acceleration. Likewise, we incorporated typical ERM activation latency into the temporal parameters of our implementation and observed no effect on perceived flow quality.

Thermal Actuator. A single Peltier-based thermal actuator (TEGWAY S017A026026) is mounted adjacent to the actuator array, positioned 1 cm off-grid on the side most appropriate for the desired path direction. This layout enables the system to exploit vibration-induced thermal referral, allowing the perceived source of temperature to be shifted onto the grid via tactually induced illusions. As the thermal source heats or cools, synchronized tactile cues guide the user’s perception of movement, enabling temperature to feel as if it is flowing across the skin, even though the heat is physically generated at a single location.

The thermal actuator, capable of a maximum temperature differential of 64°C at a voltage of 2.2 V and a current of 6.0 A, is powered and controlled using an H-bridge circuit that enables bidirectional polarity switching, allowing seamless transitions between heating and cooling phases. The actuator reaches target temperatures, mapped to input voltages via a thermal camera, in approximately 1 second and returns to base temperature in about 15 seconds. A programmable power supply (Korad KD6005P) regulates current and voltage to ensure stable operation, and a small heatsink attached with thermal tape aids heat dissipation during longer interactions. This minimalist setup supports dynamic direction changes while maintaining perceptual realism, eliminating the need for multiple distributed heat sources.

Electronic Control and Power Management. The system is orchestrated by a Raspberry Pi 4 Model B, which drives the actuator grid with motion parameters generated by a host PC and communicated over TCP. PWM signals are generated via GPIO to control individual vibration patterns, while the thermal actuator’s polarity

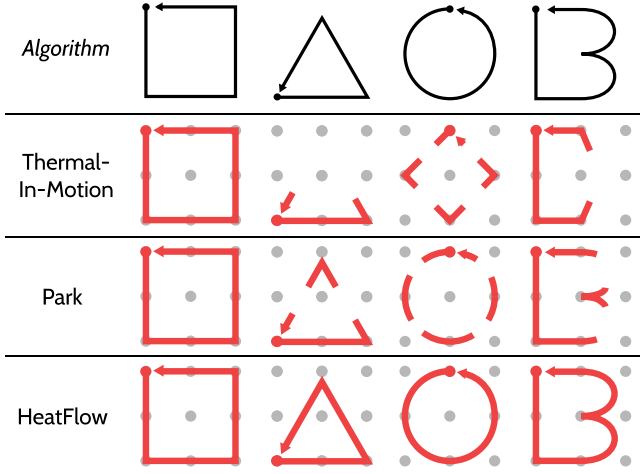


Figure 3: A diagram comparing the perception of thermal flows for various shapes when generated by 3 approaches: Thermal-In-Motion, Park algorithm, and HeatFlow.

and intensity are managed by the H-bridge and power supply configuration. This setup enables real-time responsiveness and tight synchronization between tactile motion and thermal propagation.

Customizability and Modularity. The modular housing design supports customizable array layouts, allowing different actuator configurations and density levels for various experimental or application needs. Whether creating compact displays for the palm or wraparound structures for the forearm, the HeatFlow platform can be readily adapted. This adaptability makes it an ideal candidate for deployment across wearables, handheld interfaces, and VR accessories.

3.3 Algorithm for 2D Thermal Flow Generation

HeatFlow is driven by a novel thermal-tactile control algorithm that enables expressive 2D thermal motion using a single thermal actuator and a 2D grid of vibrotactile actuators. By synchronizing tactile sequences with thermal propagation, the system creates a compelling illusion of heat or cold flowing along arbitrary 2D paths on the skin. This section presents the motivation for this algorithm, grounded in limitations of prior work, followed by details of its implementation.

Design Motivation and Limitations of Prior Approaches. Prior work has demonstrated the feasibility of simulating thermal motion through combinations of vibration and heat. However, these methods are limited in key dimensions: spatial expressivity, perceptual realism, and modality integration.

Prior Work 1: Thermal In Motion. Thermal In Motion [45] explores the potential of directional thermal illusions using a single thermal actuator and sequential tactile stimulation. By exploiting vibration-induced referral, it creates the perception of heat moving along a 1D path. While elegant in its simplicity, this approach is fundamentally limited to straight-line motion. When extended to 2D, it fails to maintain perceptual coherence due to the slowness and

spatial imprecision of thermal diffusion. Fragmented sensations and ambiguous directionality result, especially in curved or intersecting paths.

Prior Work 2: Tactile Brush. Tactile Brush [22] simulates apparent tactile motion by interpolating intensity between actuators on a regular grid. While effective for linear tactile illusions, it is constrained to grid-aligned motion, resulting in noticeable artifacts when attempting to render curved or diagonal trajectories. Moreover, it lacks support for intensity blending in overlapping paths, limiting its application to more expressive or naturalistic 2D motion — particularly when rendering continuous flow illusions.

Prior Work 3: Park Algorithm. The Park algorithm [38] improves spatial continuity by allowing phantom points to fall off-grid and blending actuator intensities using inverse-distance weighting. This allows for smoother curved tactile paths. However, it is designed solely for tactile rendering. It does not account for the temporal lag of heat perception, nor does it incorporate mechanisms for thermal referral or masking. When applied to simulate thermal motion with a single source, it leads to ambiguous flow illusions, fails to compensate for heat latency, and introduces unnecessary actuator complexity.

These limitations reveal a critical gap: existing approaches either cannot render realistic 2D thermal motion, or they fail to consider the cross-modal perceptual dynamics that govern how humans feel heat and vibration together. They often assume thermal perception behaves like tactile perception — an assumption that leads to perceptual fragmentation and technical inefficiencies, as shown in Figure 3.

HeatFlow’s algorithm directly addresses these issues. It introduces a perceptually grounded framework for 2D thermal motion that:

- Decouples tactile motion from thermal location,
- Leverages vibration-induced thermal referral,
- Enables off-grid phantom sampling, and
- Precisely modulates timing and intensity across modalities.

This allows us to render expressive thermal flows — curves, spirals, waves — using only a single thermal actuator and minimal tactile hardware.

3.4 Implementation Details and Flow Coordination

Phantom Point Selection. The algorithm begins by sampling points for phantom sensations along the user-defined 2D path at regular spatial intervals as shown in Figure 2 (a). These points define where a user should perceive motion over time. Unlike previous approaches that limit phantom points to grid intersections (e.g., Tactile Brush), we allow off-grid phantom placement to enable greater spatial resolution and flexibility, which is critical for rendering slower, more perceptually stable thermal flows.

Tactile Intensity Interpolation. Since phantom points rarely align with physical actuator locations, we interpolate vibration intensity among nearby actuators using a proximity-based blending method (see Figure 2 (b)):

$$A_{i,j} = P_k \times \sqrt{\frac{1/d_{i,j}}{D}}, \quad \text{where } D = \sum_{\substack{i,j \\ d_{i,j} < d_T}} \frac{1}{d_{i,j}} \quad (1)$$

Here, $A_{i,j}$ denotes the intensity of a tactile actuator $a_{i,j}$ in the i^{th} row and j^{th} column of the tactile grid. P_k is the phantom point intensity of phantom point p_k , which is the k^{th} phantom point along the flow path. $d_{i,j}$ is the distance from the phantom point to actuator. The constant d_T is the distance threshold that limits which actuators can contribute. This method produces a smooth, continuous sensation of motion that passes through off-grid locations.

Stimulus Onset and Timing. Each phantom point is assigned a stimulus onset time, based on a constant or curvature-adaptive stimulus onset asynchrony (SOA). This controls the pacing of the apparent motion. Faster flows result in shorter intervals between successive phantom points, while slower flows lengthen the temporal gaps, maintaining perceptual continuity.

Handling Overlapping Actuators. In some cases, multiple phantom points may influence the same actuator simultaneously, as shown in Figure 2 (c). To resolve these overlaps, we compute the final intensity $A_{i,j}^*$ of actuator $a_{i,j}$ as the average of all contributions mapped to that actuator:

$$A_{i,j}^* = \frac{\sum S_{i,j}}{|S_{i,j}|}, \quad \text{where } S_{i,j} = \{A_{i,j} \mid a_{i,j} \text{ contributes to } p_k\} \quad (2)$$

This averaging approach ensures a smooth intensity transition (see Figure 2 (d)) and avoids flickering or saturation at shared actuators, particularly during flow convergence, path intersections, or spirals.

Thermal Stimulation Timing. The thermal actuator is activated once at the start of the motion and held at the target temperature (heating or cooling) for the full duration of the path. Because thermal perception has a delayed onset and diffuses spatially, the vibration pattern is designed to lead the thermal cue. This creates the illusion that the temperature is moving along the path — even though the heat is generated from a fixed location. Tactile feedback serves as a perceptual anchor and spatial guide, helping the user track the motion of temperature across the body.

Thermal Refinement. To sharpen perceived thermal quality, we implement a warm-up period and thermal boosting technique. Before the start of a thermal flow, we prematurely activate the thermal actuator and first phantom point to allow the Peltier modules to reach target temperature and establish thermal referral [45]. Through preliminary testing, we found a 1 second duration is sufficient for warm-up before information-rich 2D flows. We also found perceived thermal intensity to intermittently decrease in off-grid locations. Thus, we slightly boost the voltage supplied to the thermal actuator for off-grid phantom points to maintain the perception of thermal evenness. To suppress unintended heat diffusion or reinforce directional cues, we optionally activate flanking actuators as masking points. These temporally and spatially structured cues

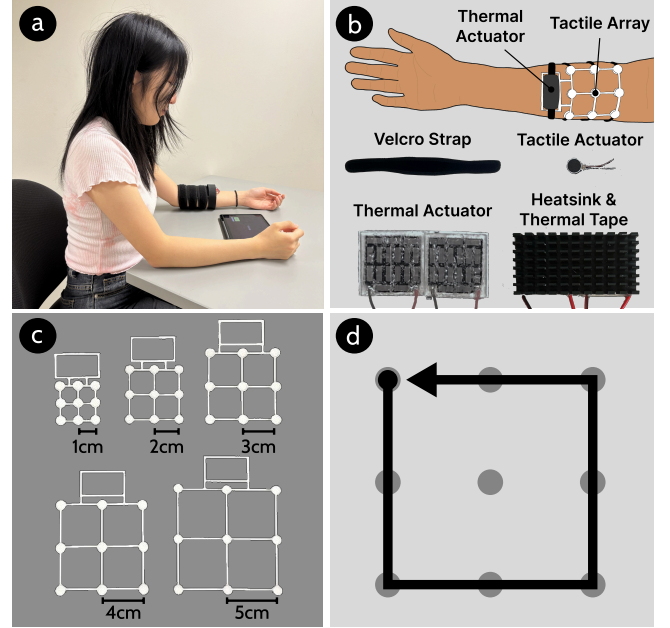


Figure 4: Depictions of the setup for user studies, displaying (a) a participant during the study, (b) the actuators and Velcro strap utilized on the forearm, (c) the grids of inter-actuator distance ranging from 1cm to 5cm, and (d) the square flow pattern used in Study 1.

reduce perceptual ambiguity, particularly for curves or bidirectional movements.

Summary. In contrast to prior work, HeatFlow’s algorithm introduces a perceptually grounded strategy for simulating 2D thermal motion. It synchronizes tactile guidance with thermal latency, supports flexible and expressive motion paths, and achieves convincing thermal illusions with minimal hardware. This enables new interaction techniques for wearable computing, virtual environments, and multisensory feedback systems that require realistic temperature dynamics across the skin.

4 User Study 1: Optimal Configuration for Thermal Flow Perception

To determine the optimal actuator spacing and motion speed for producing continuous 2D thermal flow illusions, we conducted a controlled perceptual study. Heat propagates slowly and diffusely compared to tactile feedback, making it essential to identify configurations that support both perceptual continuity and directional realism. This study investigates how inter-actuator distance, flow speed, and thermal condition influence users’ ability to perceive thermal motion.

4.1 Participants

Sixteen participants, with a mean age of 24.1 years (SD = 2.85), including eight females, took part in the experiment. All reported normal tactile and thermal sensitivity and had no prior experience with thermal-tactile interfaces. One participant reported left-hand

dominance. Participants provided informed consent and received a \$10 gift card for their participation. This study was approved by the Institutional Review Board (IRB) of the University of Texas at Dallas (IRB-21-194).

4.2 Study Design

We designed a three-factor within-subjects study to identify the optimal parameters for perceiving continuous and directional thermal flows under varying spatial, thermal, and temporal conditions. The study manipulated three independent variables: inter-actuator distance, flow speed, and temperature condition.

Inter-Actuator Distance. Five spacing conditions were tested: 1cm, 2cm, 3cm, 4cm, and 5cm. These values reflect spatial resolutions common in wearable haptic systems. Smaller distances were expected to produce more spatial overlap or blending, while larger distances could introduce perceptual discontinuity or motion gaps.

Flow Speed. Five motion speeds were selected: 10mm/s, 30mm/s, 50mm/s, 70mm/s, and 90 mm/s. These values were based on preliminary observations indicating that thermal motion is perceptible in the 1–100 mm/s range. Slower speeds allow more time for heat propagation, while faster speeds test the upper limit of thermal-tactile synchronization.

Temperature Condition. Three thermal conditions were included: *hot* (+9°C above skin temperature), *cold* (−15°C below skin temperature), and *neutral* (0°C, no thermal stimulation). These temperature values were selected based on prior work showing that strong contrasts enhance thermal motion illusions [45]. Each participant's baseline skin temperature was recorded using a thermal imaging camera (Optris PI450) before the session. To ensure participant safety, all *hot* condition temperatures were kept within the safe range established in previous studies [9]. For example, the maximum temperature applied during the study was approximately 42°C, based on a participant with a baseline skin temperature of 32.8°C. Additionally, our velcro-based apparatus allowed for quick detachment in case of discomfort. We implemented voltage limits and kill switches in the hardware, and participants were encouraged to request adjustments, take breaks, or end the session early if needed.

Flow Path and Trial Structure. In each condition, a square thermal flow path (see Fig. 4 (d)) was rendered across a 3 × 3 actuator grid. The flow began at the top-left corner and proceeded in a counterclockwise direction, traversing each edge. This setup produced 75 unique conditions (5 distances × 3 temperatures × 5 speeds). Each condition was repeated twice, totaling 150 trials per participant. Trials were grouped into five blocks based on distance. Within each block, conditions were randomized, and block order was counterbalanced across participants to reduce fatigue and order effects.

4.3 Apparatus

We prepared five modular 3 × 3 tactile grids, each consisting of nine vibrotactile actuators (JIEYI, JYC1027) arranged in a square layout. These grids were pre-configured with fixed inter-actuator distances of 1cm, 2cm, 3cm, 4cm, or 5cm (see Figure 4(c)). We position the

grids on the participant's ventral forearm, a body location that strikes a balance between surface evenness and thermal sensitivity. Each grid was 3D-printed using Acrylonitrile Butadiene Styrene (ABS), chosen for its flexibility and durability. This allowed the array to comfortably wrap around participants' forearms without losing shape or causing discomfort during repeated trials.

A Peltier-based thermal actuator was mounted in a slot positioned 1cm outside the top-left edge of the tactile grid. This placement was informed by preliminary testing, which showed that an off-grid position provided a stronger and more consistent thermal illusion while keeping the hardware compact and symmetric. The thermal actuator remained fixed throughout the experiment and served as the sole source of heat or cold across all conditions. Figure 4 (b) shows these hardware components.

4.4 Procedure

Each session began with a brief introduction, during which participants were informed of the study goals and the types of sensations they might experience. After providing informed consent, participants were seated comfortably at a desk and asked to rest their non-dominant forearm on a padded support. The HeatFlow grid was affixed securely to the forearm (see Figure 4 (a)), and a five-minute acclimation period was used to allow skin temperature to stabilize. Baseline skin temperature was recorded using a thermal imaging camera (Optris PI450) before starting the trials.

Participants completed a short training phase consisting of four sample trials to familiarize themselves with the sensations and the response format. During training, participants were encouraged to verbally describe what they felt and to ask questions to clarify their understanding of the motion illusion. For all experimental trials, participants were instructed to close their eyes to eliminate visual bias and focus on the tactile and thermal sensations.

The experiment followed a within-subjects design, where each participant completed 150 trials across five blocks. For each trial, the system first activated the thermal actuator and allowed it to reach the target temperature. A 4-second thermal-tactile flow was then rendered along a square path across the actuator grid. After each trial, participants responded to three questions:

- (1) **Q1:** Did you perceive any motion? (Yes / No)
- (2) **Q2:** What type of sensation did you feel? (Hot / Cold / Vibration / None)
- (3) **Q3:** How continuous did the motion feel? (7-point Likert scale; 1 = very fragmented, 7 = very continuous)

Responses were recorded using a tablet interface. To prevent thermal adaptation and reduce fatigue, a 10-second rest interval was included between each trial. After each distance block, participants were given a 60-second break, during which the tactile grid was replaced with the next configuration. The full session, including setup and debriefing, lasted approximately 50–60 minutes per participant.

4.5 Results and Analysis

We analyzed responses from 16 participants across 150 trials each (16 × 150 = 2,400 total trials). For each condition, we computed the percentage of trials in which participants perceived motion, the reported sensation type, and the average continuity rating on a

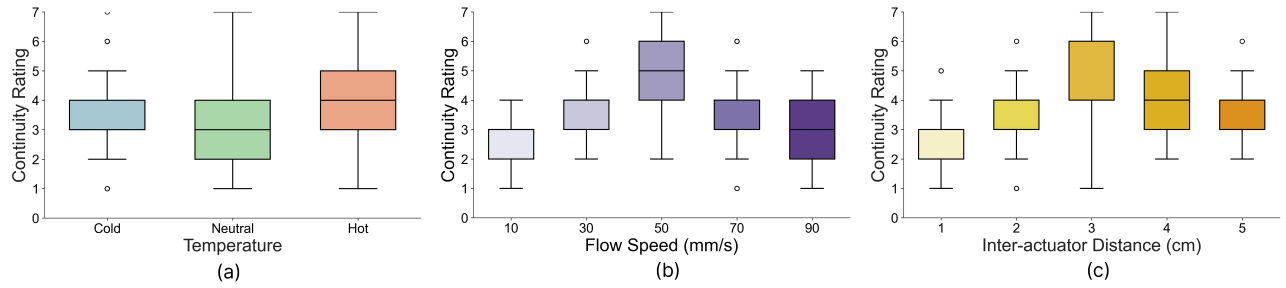


Figure 5: Mean continuity ratings of sixteen participants for (a) Temperature, (b) Flow Speed, and (c) Inter-actuator Distance. Participants reported perceiving the most convincing 2D thermal flow with a 3 cm grid spacing, 50 mm/s flow speed, and hot temperatures. Error bars represent the standard deviation of the mean.

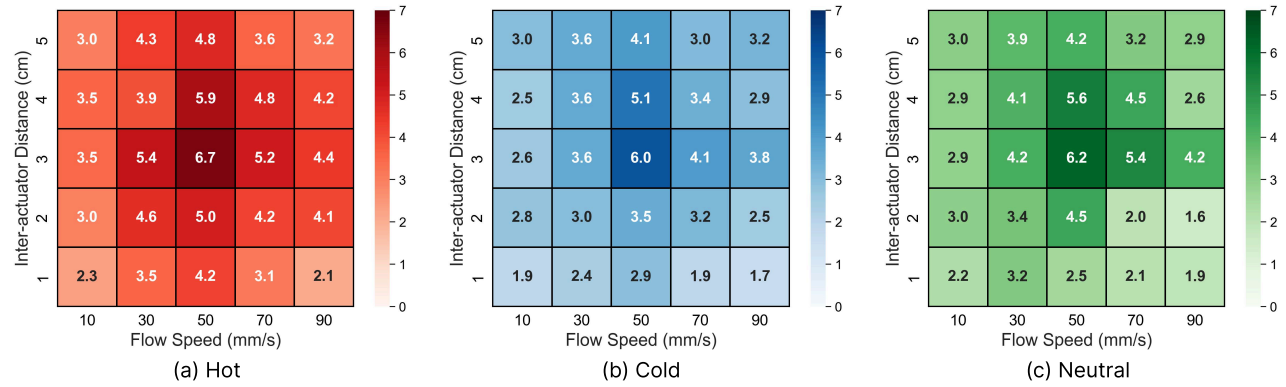


Figure 6: Heatmaps showing the mean continuity score for all combinations of flow speed and inter-actuator distance for (a) hot, (b) cold, and (c) neutral temperature conditions.

7-point Likert scale. Figure 5 presents the mean continuity ratings across all three factors, while Figure 6 illustrates the relationships between all factor combinations using a heatmap. The results show that participants perceived the most convincing 2D thermal flows when the inter-actuator distance was 3cm and the flow speed was 50mm/s. Additionally, flow perception was slightly better for hot temperatures compared to cold and neutral conditions. Data were analyzed using a Generalized Linear Mixed Model (GLMM) with a Gaussian distribution, treating temperature, inter-actuator distance, and motion speed as fixed effects, and participant ID as a random effect. Below, we report key trends and significant results for each manipulated variable.

Effect of Inter-Actuator Distance. Thermal flow illusions were most successfully perceived at 3 cm spacing, which yielded the highest motion perception rate and the highest continuity ratings ($M = 6.1$, $SD = 0.7$). The GLMM revealed a significant main effect of distance ($\chi^2 = 6.066$, $p = 0.048$). Post-hoc comparisons showed that 3cm significantly outperformed 1cm ($p < 0.001$), 2cm ($p = 0.023$), and 5cm ($p = 0.041$), but was not significantly different from 4cm ($p = 0.311$).

At 1cm and 2cm, participants reported a dense buzzing or "localized blur," likely due to excessive overlap between actuators.

Conversely, at 4cm and 5cm, participants frequently reported fragmented or "steppy" sensations, and continuity ratings dropped significantly (e.g., 5cm: $M = 3.8$, $SD = 1.2$). These results suggest that 3cm provides a perceptual sweet spot between smooth motion and clear spatial resolution.

Effect of Flow Speed. Perceived motion clarity was strongly influenced by speed. The highest continuity ratings were observed at 50mm/s ($M = 6.3$, $SD = 0.6$), followed by 30mm/s ($M = 3.78$, $SD = 0.86$) and 70mm/s ($M = 3.51$, $SD = 1.26$). The lowest scores were at 10mm/s ($M = 2.77$, $SD = 0.71$) and 90mm/s ($M = 3.01$, $SD = 1.09$). The GLMM showed a significant main effect of speed ($\chi^2 = 8.98$, $p = 0.036$), and post-hoc analysis indicated that 30mm/s differed significantly from the other speeds ($p < 0.05$).

At the slowest speed (10mm/s), participants often failed to perceive motion, describing it as "flickering" or "random pulses." At the fastest speed (90mm/s), participants noticed a thermal trail but struggled to track its direction, citing disjointed or lagging sensations. These results suggest that moderate speeds offer optimal temporal resolution for thermal motion.

Effect of Temperature Condition. Both *hot* and *cold* conditions yielded reliable motion perception across the 30–70mm/s range. However, hot temperatures produced the highest continuity ratings ($M = 4.07$, $SD = 1.23$), followed by neutral ($M = 3.43$, $SD = 1.31$)

and cold ($M = 3.20$, $SD = 1.08$). The GLMM showed a significant main effect of temperature ($\chi^2 = 61.2$, $p < 0.001$), and all interaction effects with temperature were also significant ($p < 0.001$).

In contrast, the neutral condition (no thermal input) produced significantly lower continuity ratings ($M = 3.2$, $SD = 1.4$), and several participants described the sensation as “just vibration” or “directionless.” These results confirm that thermal cues—especially warm ones—enhance the clarity and continuity of perceived motion.

To directly assess whether participants experienced perceptual fusion of thermal and tactile information, we explicitly included response options in our questionnaires that distinguished between thermal-tactile integration and vibration-only sensations. In Q2, participants were instructed to choose from ‘Hot’, ‘Cold’, or ‘Vibration’ to describe the nature of the flow they perceived. ‘Vibration’ was selected if they felt a tactile motion without thermal involvement, while ‘Hot’ or ‘Cold’ indicated that thermal and tactile cues were fused into a single, unified moving sensation. The significantly higher continuity and match ratings in Hot conditions—along with participants’ frequent selection of thermal labels over ‘Vibration’—suggest that participants did not merely feel vibration traveling along the skin, but rather experienced a coherent, fused thermal-tactile flow.

Interaction Effects. The interaction between distance and speed was particularly notable. Even at 3cm spacing, flow illusions broke down at 10mm/s due to thermal lag. Similarly, wider spacing (e.g., 5cm) underperformed at all speeds. Temperature effects were amplified at moderate speeds and optimal spacing, supporting the need for coordinated spatial-temporal-thermal control in thermal-tactile interfaces.

Summary. Together, the results indicate that a configuration of 3cm inter-actuator distance and 50mm/s flow speed, combined with either hot or cold thermal cues, yields the most perceptually convincing 2D thermal motion. These optimal parameters were adopted for all subsequent experiments, immersive demos, and the default configuration of the HeatFlow authoring tool.

5 User Study 2: Perceptual Continuity of Arbitrary Thermal Paths

Thermal sensations in real-world applications—such as simulating the movement of a heat source, tracing body contours, or mimicking expressive gestures—often follow complex, non-linear, and dynamic trajectories. These interactions demand thermal rendering techniques that can produce smooth, continuous motion across a variety of shapes, including curves, intersections, and symbolic forms. To address this challenge, we developed a novel algorithm capable of generating realistic 2D thermal flows along arbitrary paths using a combination of phantom point interpolation, spatiotemporal modulation, and masking-aware sequencing.

This study evaluates the algorithm’s performance across a diverse set of trajectories, including both geometric and expressive forms, where direction changes, curvature, and non-uniform pacing pose greater perceptual challenges. We test the algorithm’s ability to maintain motion continuity under these conditions and compare its effectiveness against alternative thermal motion rendering strategies to assess its robustness and generalizability.

5.1 Participants

Sixteen participants (8 female; $M = 22.1$ years, $SD = 2.81$) took part in the study. None had participated in the previous experiment. One participant identified as left-hand dominant; all others were right-hand dominant. No participants reported any sensory or neurological disorders affecting their hands or arms. Each participant provided written informed consent and received a \$10 gift card as compensation. This study was approved by the Institutional Review Board (IRB) of the University of Texas at Dallas (IRB-21-194).

5.2 Study Design

We conducted a three-factor within-subjects study to evaluate the effectiveness of our algorithm in rendering perceptually continuous thermal motion across diverse 2D trajectories. The study varied thermal path patterns, temperature conditions, and rendering algorithms, allowing us to assess both the generalizability and perceptual advantages of HeatFlow in comparison to alternative methods.

Rendering Algorithms. We tested three rendering algorithms:

- **Baseline 1 (Thermal-in-Motion):** A path-following strategy with fixed steps [45].
- **Baseline 2 (Park algorithm):** A radial activation technique inspired by tactile brush illusions [38].
- **Ours (HeatFlow):** A perceptually tuned algorithm using phantom point interpolation, temporal modulation, and masking-aware sequencing.

Each pattern was rendered under three temperature conditions—Hot ($+9^\circ\text{C}$), Cold (-15°C), and Neutral (0°C)—relative to each participant’s skin baseline, which was measured using a thermal imaging camera.

Path Patterns. We selected ten path patterns that span a broad range of spatial characteristics, including directionality, curvature, and complexity (see Figure 7). These included:

- **6 Geometric patterns (3 pairs of curved vs. non-curved):** Open Box, C-shape; Triangle, Semi-Circle; Diamond, Circle.
- **4 Expressive, symbolic patterns:** Spiral, Heart, Figure Eight, and Star.

This diversity allowed us to assess perceptual clarity and continuity in both structured and naturalistic motion paths.

Design Summary. The study used a full factorial design with 90 unique conditions ($3 \text{ temperatures} \times 3 \text{ rendering algorithms} \times 10 \text{ patterns}$). Each condition was repeated twice, yielding a total of 180 trials per participant. Trials were organized into 10 blocks based on the path pattern. Within each block, trials were randomized, and the block order was counterbalanced across participants using a Latin square design to minimize ordering effects.

Post-trial Responses. After each trial, participants answered three questions:

- (1) **Q1:** Did you perceive motion? (Yes / No)
- (2) **Q2:** What type of sensation did you feel? (Hot / Cold / Vibration)
- (3) **Q3:** How well the sensation matched the displayed shape, rated on a 4-point scale (1 = no match, 2 = weak match, 3 = close match, 4 = perfect match).

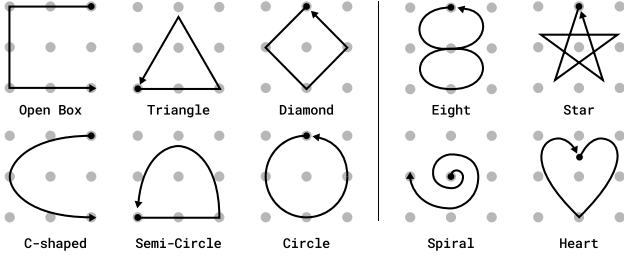


Figure 7: Figure depicting all ten thermal flow patterns used in Study 2. The left side shows the 3 pairs of curved and non-curved flows, while the right side illustrates the four commonly used symbols representing the flows.

We selected a 4-point scale without a neutral midpoint to encourage participants to make a decisive judgment about how well the rendered sensation matched the intended shape.

5.3 Apparatus

We used a setup and control system similar to 4.3 for this study but selected only the tactile grid with 3cm inter-actuator spacing, as per the results of Study 1. Following the steps described in 4.3, we situated each participant and calibrated the control system. The rendering software synchronized actuator activation in real time according to the assigned pattern, temperature, and algorithm condition.

5.4 Procedure

Each participant completed the study in a single session lasting approximately one hour. Before the trials began, participants were briefed on the study goals and provided written informed consent. We then measured their baseline skin temperature to calibrate the hot and cold stimulus levels.

The 180 trials were divided into 10 blocks, one for each path pattern. At the start of each trial, the system rendered a thermal path using a randomly assigned temperature and algorithm condition. The visual pattern corresponding to the upcoming tactile flow was displayed on the tablet 3 seconds before the trial, allowing participants to compare the sensation on their arm with the visual representation. The motion speed was fixed at 50mm/s, based on the optimal speed identified in Study 1, resulting in durations ranging from 3 to 6 seconds depending on the path length. Following each stimulus, participants answered three questions about motion perception, sensation type, and perceived continuity. All responses were recorded on a tablet. A 10-second inter-trial interval allowed skin temperature to return to baseline.

Participants were instructed to remain still and relaxed throughout the session. A 60-second break was provided after each block to minimize fatigue. After completing the final block, participants filled out a brief post-experiment questionnaire capturing their subjective impressions of the thermal flows, including comparisons of different temperatures and path patterns.

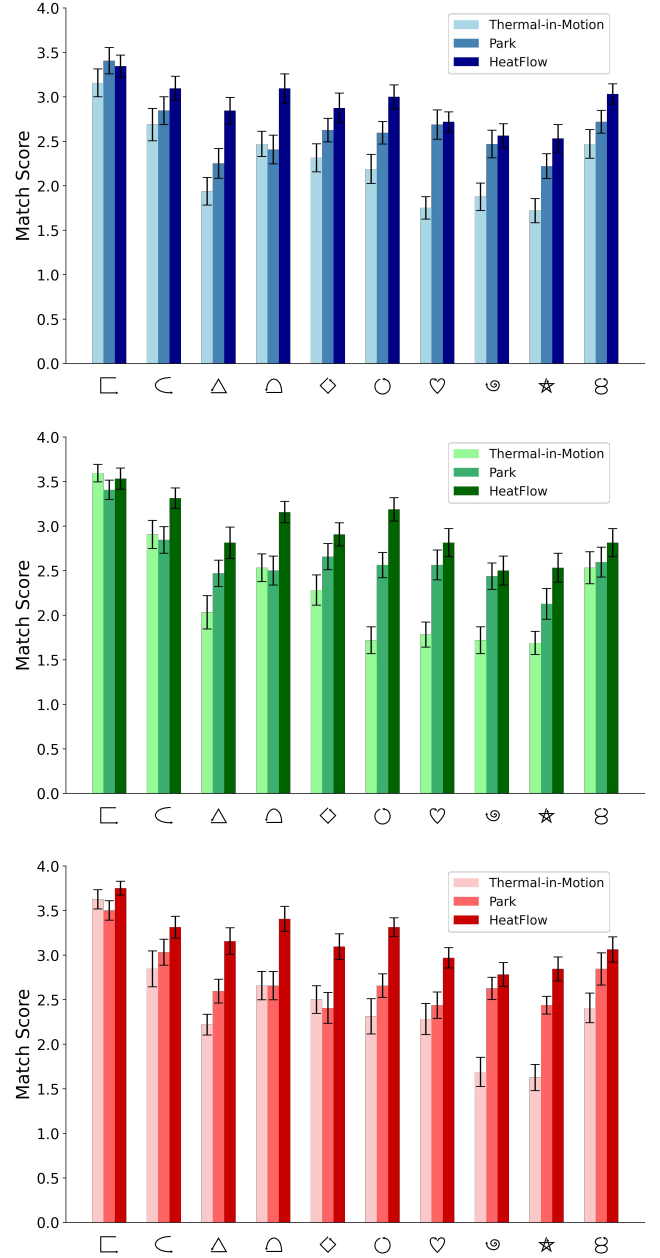


Figure 8: The mean match score for all ten flow patterns comparing the performance of HeatFlow algorithm with the two baselines for cold (blue), neutral (green), and hot (red) temperatures. Results suggest that HeatFlow algorithm clearly outperformed both the baselines in rendering most of the patterns.

5.5 Results and Analysis

We analyzed a total of 2,880 trials (16 participants $\times$ 180 trials), evaluating motion perception, sensation type, and perceived shape recognition. For each trial, we computed the percentage of perceived motion and the average match score on a 4-point scale, where

higher values indicated closer alignment with the displayed thermal pattern.

We confirmed non-normality using the Shapiro-Wilk test ($p < 0.001$) and unequal variances via Levene’s test ($p < 0.001$). We, therefore, used a Generalized Linear Mixed Model (GLMM) with a Gaussian family and identity-link function, treating algorithm, temperature, and pattern as fixed effects and participant as a random effect.

Algorithm Comparison. A GLMM revealed a significant main effect of algorithm on match scores ($\chi^2 = 26.34$, $p < 0.001$). Ours (HeatFlow) achieved the highest overall score ($M = 3.05$), significantly outperforming both Baseline 2 (Park) ($M = 2.64$) and Baseline 1 (Thermal-in-Motion) ($M = 2.37$). Pairwise comparisons showed these differences were statistically significant ($p < 0.01$).

HeatFlow’s advantage remained consistent across all temperature conditions. In the hot condition, it achieved an average match score of 3.19, compared to 2.44 for Baseline 1 (Thermal-in-Motion) and 2.64 for Baseline 2 (Park). In the neutral condition, HeatFlow scored 2.97, while Baseline 1 (Thermal-in-Motion) and Baseline 2 (Park) dropped to 2.09 and 2.56, respectively. In the cold condition, HeatFlow maintained a strong performance with a score of 2.84, outperforming Baseline 1 (Thermal-in-Motion) ($M = 2.26$) and Baseline 2 (Park) ($M = 2.55$). These results demonstrate that HeatFlow provides a stable advantage across diverse thermal stimuli, enabling smoother and more recognizable 2D thermal motion patterns.

Pattern Shape Effects. Performance varied across patterns, with higher match scores observed for geometric shapes like *C-shape* and *Open Box* ($M = 3.43$). More complex and curved paths—such as *Heart* and *Spiral*—were generally more challenging for all algorithms. However, HeatFlow still maintained an edge: it achieved scores of 2.83 for *Heart* and 2.61 for *Spiral*, while both baselines scored below 2.0 on these patterns.

In direct comparisons between paired patterns (e.g., *Circle* vs. *Diamond*, *C-shape* vs. *Open Box*), HeatFlow preserved performance better than the other two algorithms. Baseline 1 (Thermal-in-Motion) and Baseline 2 (Park) showed noticeable drops in curved or closed-loop shapes, highlighting their vulnerability to directional shifts and path complexity. HeatFlow, in contrast, demonstrated greater flexibility and spatial adaptation.

Subjective Impressions. Participant feedback aligned with the quantitative trends. HeatFlow was often described as surprising and engaging, with many impressed by the ability to perceive shapes through thermal and vibrations. Larger, smoother curves (e.g., *Circle*, *C-shape*) were easier to follow and felt continuous, while smaller or more complex shapes (e.g., *Heart*, *Spiral*, *Eight*) were harder to track, especially where strokes overlapped. Straight shapes were typically clearer, particularly when aligned with the actuator grid.

6 User Study 3: VR Experiences

VR presents a compelling context for multisensory feedback, where thermal cues can enhance realism, emotion, and immersion. To evaluate HeatFlow’s effectiveness in such applications, we designed a user study involving six immersive VR scenarios, each representing a fixed combination of body site and thermal narrative. This study examined whether dynamic 2D thermal flows delivered through

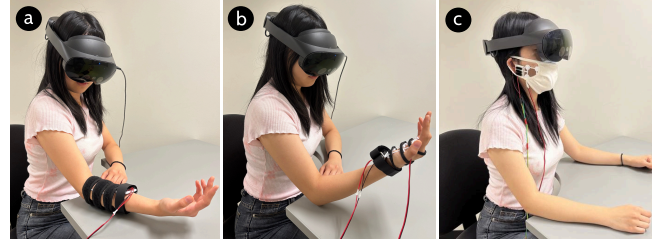


Figure 9: A user wearing HeatFlow on the (a) forearm, (b) palm, and (c) face.

HeatFlow could improve users’ immersion, enjoyment, and satisfaction compared to a non-thermal tactile baseline. We explored how HeatFlow performs when embedded in rich, interactive environments—across the forearm, palm, and face—where the spatial and affective qualities of temperature play an essential role in shaping virtual experiences.

6.1 Participants

Sixteen participants (mean age = 21.8 years, $SD = 2.21$; 8 females), who had not previously participated in Study 1 or Study 2, took part in the experiment. All were right-handed and reported no sensory impairments affecting cutaneous perception. Six participants had prior experience with VR, and two reported familiarity with haptic technologies. Each participant provided written informed consent and received a \$10 gift card. This study was approved by the Institutional Review Board (IRB) of the University of Texas at Dallas (IRB-21-194).

6.2 Setup

Participants wore a Meta Quest 3 VR headset connected to a host PC (Intel Core i7 CPU, 32GB RAM, NVIDIA GTX 1650 GPU) running the experimental software. HeatFlow actuator arrays were mounted on three distinct body sites—forearm, palm, and face—each using a customized 2D configuration optimized for that location (see Figure 10):

- **Forearm:** A 3×3 actuator grid with 3cm spacing, wrapped using elastic Velcro straps.
- **Palm:** A 3×3 actuator grid with 2cm spacing for male participants and 1.5cm for female participants, mounted on a semi-rigid backing resting in the open palm.
- **Face:** Two lightweight 3×2 actuator grid with 1.5cm spacing positioned along the cheekbone and jawline, mounted onto an off-the-shelf face mask using medical-grade adhesive strips.

Each actuator grid integrated vibrotactile motors and thermal actuators to deliver directional cues using the HeatFlow algorithm. Thermal and tactile actuators were not co-located; instead, each array contained independently positioned heating/cooling elements and vibration motors placed in close proximity to coordinate thermal-tactile motion. Thermal and tactile feedback from the actuator grid was synchronized with audio-visual stimuli in the VR content.

6.3 Study Design

The experiment followed a within-subjects design with two main factors: feedback type and VR scenario. Each participant completed all twelve conditions, experiencing six distinct virtual scenarios in both the HeatFlow (thermal + tactile) and Non-Thermal (tactile-only) conditions.

The feedback type manipulation allowed us to isolate the effect of thermal actuation. In the HeatFlow condition, participants received directional thermal and vibrotactile feedback rendered through our algorithm. In the Non-Thermal condition, the same visual and auditory content was used, but thermal actuation was disabled—providing only tactile motion cues.

Each of the six VR scenarios represented a unique combination of body site (forearm, palm, or face) and thermal context (hot or cold). The pairing of scenario and thermal condition was fixed and designed to reflect a consistent narrative theme. For example, the Tattoo Parlor scene rendered heat on the forearm, while Paintbrush conveyed a cold sensation in the same area.

To support controlled comparison across body regions, trials were grouped into three blocks, each corresponding to a body site. Within each block, participants experienced both the hot and cold scenarios for that site, once in the HeatFlow condition and once in the Non-Thermal baseline. The order of blocks, as well as the feedback condition order within each block, was counterbalanced across participants using a Latin square design to mitigate potential ordering effects.

Visual content, audio, and tactile motion cues were held constant across conditions. The only difference between HeatFlow and Non-Thermal trials was the presence or absence of thermal feedback.

6.4 VR Scenario Design

To explore HeatFlow’s effectiveness in emotionally expressive and context-rich environments, we developed six narrative-based VR scenarios that paired specific body sites with corresponding thermal experiences. Each scenario was designed to evoke either a hot or cold sensation in a thematically coherent setting, emphasizing spatial thermal motion, perceptual realism, and affective engagement.

Each scenario lasted approximately five minutes and incorporated synchronized visual, auditory, and tactile components. In the HeatFlow condition, thermal motion was rendered in real time alongside vibrotactile cues using the HeatFlow algorithm. In the Non-Thermal baseline, participants experienced the same scenes with identical audiovisual and tactile elements, but without thermal actuation.

The scenarios were grouped into three sets, one for each body site—forearm, palm, and face. Within each set, one scenario conveyed a hot experience and the other a cold one. The actuator configurations used to render these sensations are described in Section 6.2.

Tattoo Parlor (Forearm, Hot). Participants entered a virtual tattoo parlor and experienced the sensation of a heart being tattooed onto their forearm (see Figure 10 (a)). The tattoo machine buzzed as a red heart shape was traced along their avatar’s arm. In the HeatFlow condition, directional vibrotactile motion was paired with

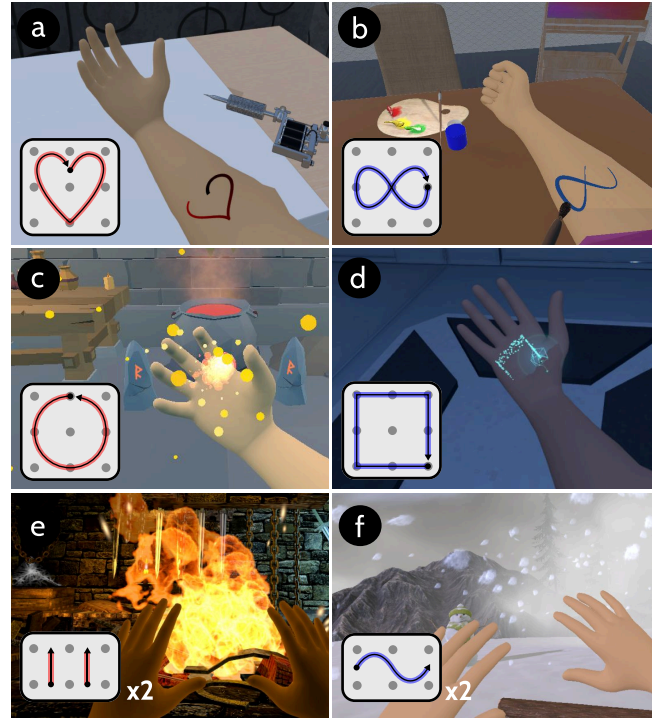


Figure 10: Figure illustrating all six VR scenarios, with the corresponding flow rendering displayed at the bottom left of each: (a) Tattoo Parlor, (b) Paintbrush, (c) Magical Fireball, (d) Sci-Fi Particles, (e) Blacksmith Forge, and (f) Snowstorm.

thermal stimulation at 38°C, simulating the warmth and localized pressure of a real tattooing process.

Paintbrush (Forearm, Cold). In this scenario, participants stood in a colorful art studio and felt a chilled paintbrush glide across their forearm in the shape of an infinity symbol (see Figure 10 (b)). The brushstroke was visually animated and accompanied by soft ambient sound. In the HeatFlow condition, vibrotactile cues were accompanied by thermal cooling at 28°C, creating the impression of cold, wet paint spreading across the skin.

Magical Fireball (Palm, Hot). Set in a fantasy dungeon, this scene allowed participants to conjure a magical fireball in their palm (see Figure 10 (c)). After the cauldron in front of them erupted, the fireball formed and began to rotate along a circular path across the hand. In the HeatFlow condition, synchronized thermal feedback at 36°C was combined with pulsing vibrotactile cues, simulating the movement and energy of growing magical heat.

Sci-Fi Particles (Palm, Cold). This futuristic scene placed participants in a high-tech laboratory where they interacted with a chilled energy stream (see Figure 10 (d)). A blue, shimmering square traced across the palm as if data or energy were flowing into the hand. In the HeatFlow condition, thermal feedback at 24°C accompanied inward vibrotactile motion, reinforcing the sensation of cold, reactive particles being transferred through the skin.

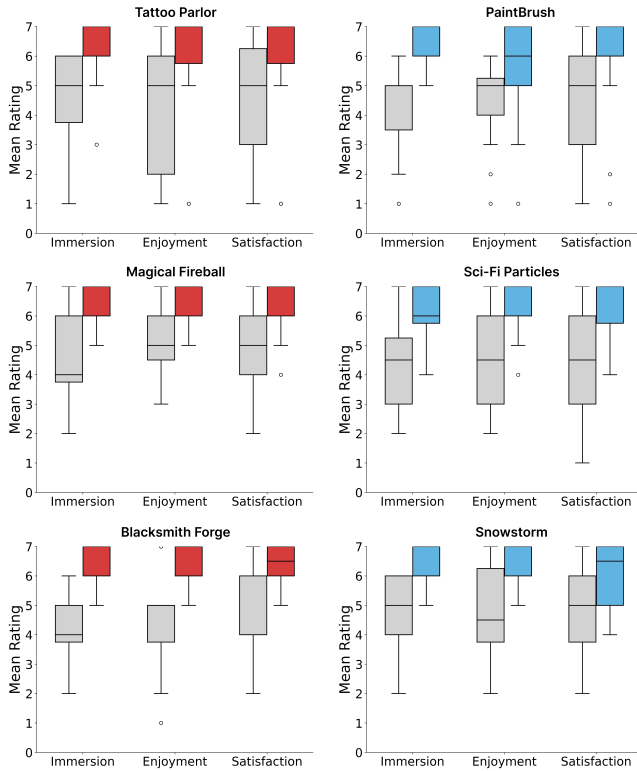


Figure 11: Comparison of Thermal and Non-Thermal conditions for Immersion, Enjoyment, and Satisfaction across all six VR scenarios. The results clearly show that the thermal flows rendered using the HeatFlow algorithm were preferred over vibration-only flows.

Blacksmith Forge (Face, Hot). Standing near a virtual forge, participants watched molten metal being shaped on an anvil (see Figure 10 (e)). Periodically, bursts of fire and sparks shot outward, sending waves of heat across their face. In the HeatFlow condition, vertical pulses of thermal stimulation at 36°C were delivered across the cheeks in sync with the hammer strikes, simulating the blast of heat from active metalwork.

Snowstorm (Face, Cold). In a snowy mountain scene, participants were immersed in a sudden blizzard (see Figure 10 (f)). Wind and snow swept across their face from front to back, driven by visual particle effects and spatial audio. In the HeatFlow condition, horizontal thermal flows at 24°C simulated the brisk sensation of cold wind moving over the cheeks, reinforcing the feeling of exposure to icy outdoor conditions.

6.5 Procedure

Participants were seated at a desk and asked to read the study instructions before signing a consent form. The experimenter explained the procedure in detail and confirmed participant understanding before placing the appropriate 2D actuator grid on the body site relevant to the upcoming scenario. Participants wore a VR headset and rested their arms comfortably during each trial.

Each virtual scene was presented twice—once with HeatFlow (thermal + tactile feedback) and once with the Non-Thermal baseline (tactile-only). The order of feedback conditions was counterbalanced across participants. After each pair of trials, participants completed a brief questionnaire using continuous 0–7 scales to compare the two feedback conditions on three dimensions: Immersion, Enjoyment, and Satisfaction. These ratings were collected after all six scenarios, resulting in twelve sets of responses per participant. Upon completing all scenarios, participants participated in a 12-minute semi-structured interview to reflect on their experiences and preferences. Each scenario lasted about five minutes, and the full session, including setup, took approximately one hour.

6.6 Results and Insights

We report both quantitative results and qualitative feedback to assess the impact of HeatFlow on immersion, enjoyment, and satisfaction in narrative-driven VR scenarios.

Quantitative Findings. Participants rated each feedback condition on a continuous 0–7 scale across Immersion, Enjoyment, and Satisfaction. As shown in 11, HeatFlow was rated significantly higher than the Non-Thermal baseline across all six scenarios: Immersion ($M = 6.39$ vs. 4.22), Enjoyment ($M = 6.03$ vs. 4.41), and Satisfaction ($M = 6.13$ vs. 4.47). A two-way repeated-measures ANOVA confirmed a significant main effect of feedback type ($p < .001$) for all measures. No significant differences emerged across scenarios, indicating the generalizability of HeatFlow’s impact.

Although not statistically significant, facial interactions received the highest average ratings, followed by the palm and then the forearm. These trends suggest that some body sites—especially those involved in social or environmental perception—may enhance thermal expressiveness in VR.

Sensory Congruity Enhances Presence. Participants widely noted how thermal and tactile cues aligned with visuals and audio to strengthen immersion. Thermal feedback was described as “natural,” “spot on,” or adding a new “dimension” to the experience. Scenarios like Snowstorm and Tattoo Parlor were praised for evoking realistic environmental or bodily sensations. While most praised the multi-sensory integration, a few noted that vibration sometimes felt mismatched or overly strong, suggesting a need for better calibration.

Thermotactile Feedback Deepens Embodiment. Directional temperature cues were frequently described as meaningful and intuitive. Heat and cold were interpreted as magical energy, wind, paint, or fire—physically anchoring participants within the scene. Many participants reported reacting instinctively to temperature changes, “feeling it move,” or recognizing familiar sensations despite knowing the effect was simulated.

Body Site Preferences. Participants expressed a range of preferences. The forearm was valued for being unobtrusive and contextually relevant; the palm was described as more comfortable and familiar. Although fewer preferred the face, several noted it enhanced emotional impact in environmental scenes. Some recommended expanding coverage or adjusting vibrational intensity to better suit facial sensitivity.

Thermal Preferences. Cold stimuli were more frequently preferred, described as comforting, refreshing, or precise. Others favored warmth, particularly in scenes where heat substituted for pain or reinforced narrative realism. These preferences highlight the importance of personalization in thermal interaction design.

Emotional and Cross-Modal Responses. Participants experienced illusions and emotional reactions that extended beyond basic perception. The Paintbrush scene commonly evoked wetness, while Tattoo Parlor elicited tension and anticipation. These effects demonstrate HeatFlow’s capacity to engage users on cognitive and affective levels, contributing to richer, more embodied storytelling.

Taken together, the quantitative and qualitative findings demonstrate that HeatFlow significantly enhances user experience by increasing realism, presence, and emotional resonance. Its multisensory integration supports immersive narrative design while enabling nuanced, personally meaningful interactions.

7 HeatFlow Authoring Tool

To broaden the impact of HeatFlow beyond research and into creative immersive media, we developed a companion design tool that enables users to craft and test their own expressive thermal flows. The interface leverages HeatFlow’s spatiotemporal capabilities, offering precise control over motion, intensity, and temperature. Users can compose, preview, and synchronize thermal effects with interactive content—such as simulating a slow build-up of heat in a dramatic cutscene or rendering rapid hot–cold transitions for dynamic gameplay moments. This tool opens new avenues for storytelling and affective design through multimodal thermal expression.

7.1 Interface Design

We implement the HeatFlow design tool through a tablet-and-pen interface (Samsung Galaxy Tab S9 FE, Samsung S Pen) to employ intuitive drawing-based input methods. The tool interface (see Figure 12 (b & c)) supports creation and customization of thermal flows, direct integration with audiovisual media content, and interactive connection with the HeatFlow display for rapid iteration of thermal flow designs. We present the following features that enable users to create diverse, multisensory experiences:

Freehand Patterns. Any 2D flow pattern a user imagines can be captured through a freehand drawing canvas. The authoring tool handles all algorithm calculations and visualizes design-relevant information. Our interface enables users to quickly create and experience unique thermal flows through the HeatFlow display.

Customizable Parameters. The tool grants users extensive command over thermal flow configuration. Users can precisely control the speed, duration, phantom point count, and temperature of their custom thermal flows. The efficiency of the HeatFlow algorithm allows rapid recalculation of flow parameters with each user action, facilitating a smooth and responsive editing experience as users refine thermal flows to full satisfaction.

Dynamic Sensations. We introduce per-point modification of tactile intensity and thermal temperature in the authoring tool to

inspire dynamically expressive thermal flows. Users can adjust the tactile and thermal intensities of each individual phantom point through intuitive pressure-sensing input. We empower user creativity through implementation of flow dynamism, anticipating artistic realizations of rich sensory experiences.

Synchronized Experiences. The HeatFlow authoring tool offers transfer and synchronization of thermal flows to digital video and virtual reality content, facilitating rapid design of immersive and interactive media. Inspired by existing video editing software, the tool interface enables users to directly link their thermal flow designs to imported audiovisual content, avoiding complex system overhead. When ready for delivery, users can connect to the HeatFlow display and seamlessly initiate the synchronized experience.

We present the HeatFlow suite to reveal new multimodal opportunities in immersive experiences. Through the HeatFlow technique, display, and design tool, anyone can augment a sensory interaction with compelling thermal flows, stimulating unique dimensions of human perception.

7.2 Evaluation

To evaluate the effectiveness and impact of HeatFlow on user experience, we invited eight new participants (4 female, mean age = 24.5 years, SD = 1.98) to explore HeatFlow’s capabilities through an interactive session with the design tool. In a lab setting, participants were introduced to the HeatFlow design tool and asked to wear a 3×3 HeatFlow grid on the non-dominant ventral forearm location as shown in Figure 12 (a). We selected the non-dominant forearm for the HeatFlow grid in this interactive session to allow unobstructed use of the dominant arm with the design tool. Next, the experimenter familiarized the participant with the HeatFlow suite through a demonstration tutorial. The participant confirmed understanding by completing simple warm-up tasks with the design tool. Once ready, we asked the participant to design thermotactile feedback for four immersive media experiences with a rich variety of thermal-related visual stimuli (see Figure 12 (d & e)):

- *Avatar*: A video from the live-action series *Avatar: The Last Airbender* with blasts of fire and wind in an icy land.
- *Panda*: A video from the movie *Kung Fu Panda 2* featuring sparkling fireballs and sprays of water.
- *Dragons*: An original VR environment made in Unity where two dragons fight with fire and ice breaths.
- *Camping*: An original VR environment made in Unity, where a lit campfire is doused by an approaching thunderstorm.

Participants were encouraged to design thermal flows based on their personal interpretation of the audiovisual content and replay the experience until satisfied. We included a video version of each VR experience in the design tool video player to help participants synchronize thermal flows with the VR action. Participants were provided a Meta Quest 3 headset to utilize at their discretion. Once all four experiences had been explored, semi-structured interviews were conducted with the participant to gather their feedback and insights.

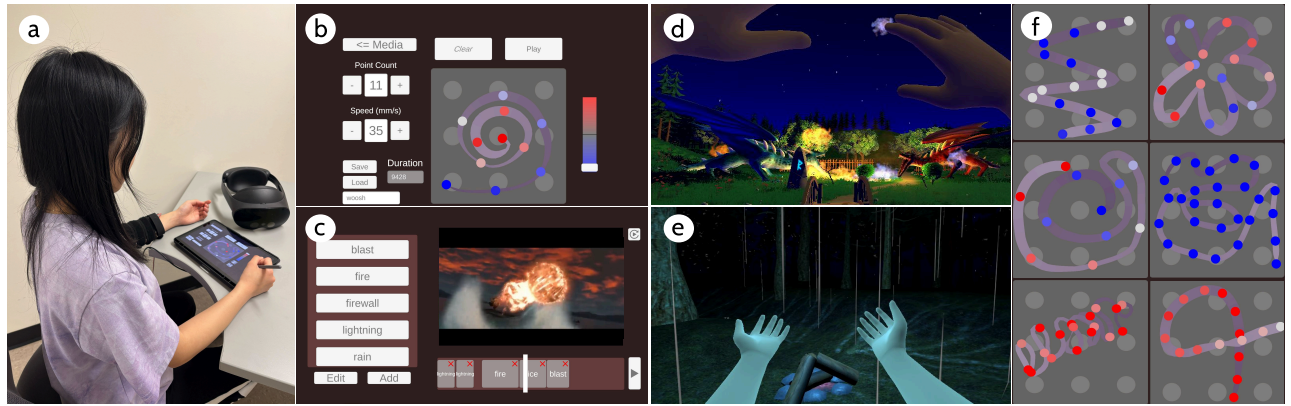


Figure 12: Figure showing: (a) a user designing a customized thermal flow, (b) authoring tool - design window, (c) authoring tool - sequencing window, (d) the Dragons VR scene, and (e) the Campfire VR scene, and (f) examples of user-created thermal flows.

7.3 Insights

Multimodal Integration Enhances Experience. Participants delighted in the multimodal engagement of their senses during the immersive experiences. Thermotactile feedback from HeatFlow was considered "exciting" and "really powerful". Many participants were "surprised to really feel" the action of visual scenes, and application in entertainment media was a common suggestion.

Dynamic Flows Convey Richer Information. Every participant explored dynamic intensity and temperature changes to replicate audiovisual content, expanding the amount of information conveyed through their custom flows. Tactile dynamism was particularly significant for the user experience in scenes of "high action", and many participants enjoyed actualizing volatile collisions of "hot and cold in one" through rapid temperature shifts. In general, participants felt the dynamic features were "essential" to their ability to express ideas.

Simplicity Empowers Expressivity. Participants felt empowered to create by the intuitive design of the tool's interface and seamless integration of media content. The design process was often described as "easy to learn" and "really straightforward." We observed several unique flow designs throughout the sessions (see Figure 12 (f)), and many participants felt capable of "making the timings match perfectly." As the relevance of multisensory feedback continues to grow in the domain of immersive media, we must continue to equip users with the user-centric tools to create such experiences.

Adaptability to the Abstract. Many participants explained unique approaches to designing thermal flows for abstract scenarios, such as "mimicking the scenario" and "associating" visual content with symbols. For example, we observed spiraling flows for wind swirls, jagged lightning bolts to simulate electricity, and sudden drops in temperature to induce emotional shock. Some participants even synchronized feedback intensity to the music of the scene. Personal interpretation of audiovisual content significantly shaped resulting flow designs, revealing compelling insights into how users perceive and translate sensory experiences.

Affective Feedback. Beyond simulating audiovisual content, participants found HeatFlow could elicit deeper human feelings and emotions. HeatFlow feedback received various descriptors, ranging from "pleasant" and "therapeutic" to "shocking" and "like goosebumps." Several participants likened thermal components to human emotions, such as "hot when you're ashamed or angry" and "burst of cold for surprise." We emphasize that thermal flows capture more than thermotactile sensations. HeatFlow is a powerful platform to design interactive stories and to express human experiences.

8 Discussion

HeatFlow demonstrates that expressive 2D thermal motion can be rendered with minimal hardware by leveraging perceptual illusions. By combining phantom point interpolation, vibration-induced thermal masking, and spatiotemporal coordination, our approach simulates continuous thermal flows using a 3×3 vibrotactile array and a single thermal actuator. This method significantly reduces hardware complexity while preserving clarity and expressiveness in the rendered feedback. Our work contributes not only a novel system but also empirical insights that advance the design of perceptually tuned thermal-tactile interfaces.

8.1 Perceptual and Technical Insights

HeatFlow demonstrates that even with minimal hardware, carefully tuned illusion-driven rendering can produce rich and continuous 2D thermal motion. We investigated various inter-actuator spacings and found that a 3 cm spacing on the forearm yielded the clearest and most continuous perception of motion. While other spacings also produced recognizable patterns, 3 cm offered the best balance between actuator density and perceptual continuity. This establishes a practical design baseline, though body-specific optimization remains an open area due to differences in curvature, skin sensitivity, and thermoreceptor distribution [26, 35].

We also found that the perception of thermal flows depends on the interplay between flow speed, temperature, and path complexity. Hot exhibited significantly higher perceptual clarity ratings compared to Neutral and Cold. The moderate 50mm/s flow

speed conditions were rated highest, and ratings decrease as speed approaches extreme values. Furthermore, complex patterns with more segments and directional changes—such as stars and figure-eights—were rated higher when generated by our algorithm as compared to prior work [22, 38, 45]. These findings suggest careful consideration of thermal perception features may enhance motion interpretation by providing more continuous sensory information [34, 37]. Designers of thermal cues should consider these variables carefully when crafting interactive feedback.

8.2 Neurophysiological Basis of HeatFlow

We hypothesize that the HeatFlow percept arises from multisensory integration processes occurring at both cortical and peripheral levels. At the cortical level, prior research has identified convergence zones—such as the posterior insular cortex and the thalamus—where neurons respond to both thermal and tactile inputs [13]. These multisensory regions likely play a role in resolving spatial and temporal discrepancies between modalities, resulting in a unified percept. In particular, the thermal referral effect—central to our approach—is thought to involve cortical-level remapping, wherein thermal sensations are localized according to concurrent tactile context.

At the peripheral level, interactions between mechanoreceptors and thermoreceptors may further contribute to the illusion. For example, the sensitivity of Pacinian corpuscles, which encode high-frequency vibration, can be modulated by temperature. Moreover, mechanoreceptive afferents have been shown to respond to rapid temperature changes, as in the temperature–weight illusion [26]. These peripheral mechanisms may help prepare the somatosensory system for cross-modal binding, especially when thermal and tactile inputs are spatially aligned. Together, these cortical and peripheral processes likely underlie the perceptual remapping and coherence observed with HeatFlow.

8.3 Cross-Modal Synchronization and User Experience

HeatFlow’s ability to deliver smooth, believable sensations stems from precise coordination between tactile and thermal stimuli. By tuning SOA, we ensured that the faster tactile cue led the slower thermal input just enough to maintain perceptual alignment. Participants frequently described sensations as “flowing,” “natural,” or “lifelike,” suggesting that the multisensory calibration process enhanced coherence and immersion [24].

This effect was particularly strong in our VR applications, where visual and auditory elements were combined with thermal-tactile flows. Participants reported heightened emotional engagement and even sensations of wetness. This perceptual amplification underscores the powerful role of multisensory synergy in VR and suggests that dynamic thermal cues can meaningfully elevate presence and realism in virtual environments. These results align with broader literature on sensory integration and immersion [19, 44], but also highlight new design opportunities for leveraging thermo-tactile illusions in digital contexts.

8.4 Design Implications and Application Domains

Our authoring tool enables designers to define expressive motion paths and parameterize thermal flow based on motion speed, direction, and intensity. The ability to render recognizable symbolic shapes—such as hearts, stars, and spirals—suggests that thermal cues can communicate not only spatial information but also affective or semantic meaning. These forms of thermal expression open doors for applications in VR storytelling, affective communication, wellness interventions, and ambient displays. One promising direction for thermal flow design is integration with existing tactile animation tools, such as Mango by Schneider et al. [41], which may reveal novel paradigms for thermo-tactile authoring.

We also demonstrated that HeatFlow can adapt to various body locations. In addition to the forearm, we tested palm-based applications and twin 3×2 actuator grids to simulate thermal flows on the face. Participants were able to discern motion across these sites, and motion parameters could be tuned to body-specific preferences. This versatility supports integration with existing haptic interfaces [14, 27, 31, 39, 55, 61, 62, 65] and the development of customizable wearable haptics, where form factors and flow designs are matched to specific anatomical or interaction needs. Future physical implementations might further explore unique somatic sensations by shaping actuator arrays to align with body contours.

8.5 Vision and Future Opportunities

While we demonstrated HeatFlow primarily within immersive VR scenarios, its ability to deliver expressive, dynamic thermal motion across the body opens a much broader design space. The unique combination of tactile timing, thermal masking, and phantom point illusion makes HeatFlow not just a haptic device, but a platform for embodied communication, creativity, and accessibility.

Expressive and Affective Communication. HeatFlow enables a new sensory language for emotional and social expression. Thermal gestures—such as a warm pulse or a cooling sweep—can convey comfort, tension, or emotional nuance during digital interactions [20]. These could serve as personal thermal “signatures” in messaging or social VR. Similarly, HeatFlow could enrich XR narratives through thermal storytelling, like rising warmth during a climactic scene or a cool spiral during eerie moments. In wearable fashion, programmable thermal flows may offer new forms of self-expression or mood display [10].

Embodied Interaction and Feedback. On-body thermal cues can serve as discreet, ambient notifications for digital events or state changes [4, 18]. A warm wave across the wrist might indicate a message; a cold tap could signal a navigation cue. Thermal directionality could also assist users with visual impairments by providing symbolic spatial feedback. These silent, private interactions suit mobile and multitasking contexts, offering alternatives to sound or vibration.

Therapeutic and Mindful Experiences. HeatFlow could support wellness through touch therapy, relaxation, or guided breathing. Dynamic thermal motion can simulate massage-like effects or be synchronized with biosignals for biofeedback [12, 33, 51]. In

rehabilitation, gamified experiences might encourage users with nerve damage to follow or recognize thermal motion as part of somatosensory retraining.

Training, Simulation, and Skill Learning. HeatFlow has potential in performance training and immersive simulation, utilizing the significance of thermal cues in human perception [28, 54]. In gaming, it could simulate environmental effects like fire or ice. In sports or music training, thermal cues could signal timing or movement changes faster than visual updates. Training scenarios for tactile precision—such as tattooing, surgery, or drawing—could benefit from the fine spatial and emotional feedback HeatFlow provides.

These application themes reflect HeatFlow’s broad utility as an expressive, wearable, and embodied interaction platform. As thermal motion becomes a design language, we anticipate new multisensory experiences that merge emotion, information, and sensation—delivered directly through the skin.

8.6 Limitations and Future Work

Although our approach demonstrates strong performance, there are important limitations to consider. First, the system is currently limited to a small display area, and scaling to larger or more complex surfaces will require rethinking actuator layout, power delivery, and heat dissipation strategies [14, 28]. Second, while our masking strategy mitigates thermal diffusion and latency, longer-term usage may introduce adaptation effects that diminish clarity. We did not explore prolonged use or sensory fatigue, which are relevant for wearable or ambient interfaces [53].

Hardware constraints remain: the current prototype is wired, rigid, and not suited for use in mobile wearables. We plan to explore flexible substrates and textile-integrated actuators [48, 57] to improve comfort and portability. Our authoring tool currently supports offline pattern creation, but future versions will support real-time, feedback-driven editing for VR designers. We also envision adaptive rendering that dynamically adjusts flow parameters based on body location, user preferences, or environmental context. These developments will broaden HeatFlow’s creative and practical potential as a multisensory interaction platform.

9 Conclusion

This work introduces HeatFlow, a thermal display designed to create dynamic thermal sensations across a 2D surface by integrating thermal and tactile stimuli. Through three studies, we gained insights into optimizing tactile array size, effective speeds for thermal motion, and ideal temperatures. We also extended our approach to simulate curved thermal flows. Our findings confirm that HeatFlow can generate continuous and dynamic thermal movements across various body locations and effectively validate this approach within virtual reality environments.

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References

- [1] David S. Alles. 1970. Information transmission by phantom sensations. *IEEE transactions on man-machine systems* 11, 1 (1970), 85–91.
- [2] Keisuke Arai, Satoshi Hashiguchi, Fumihisa Shibata, and Asako Kimura. 2017. Analysis of paradoxical phenomenon caused by presenting thermal stimulation on three spots. In *International Conference on Human-Computer Interaction*. Springer, 281–286.
- [3] Keisuke Arai, Miki Matsumuro, Satoshi Hashiguchi, Fumihisa Shibata, and Asako Kimura. 2021. Hot–cold confusion: Inverse thermal sensation when hot and cold stimuli coexist in a thermal localization task. *Perception* 50, 6 (2021), 508–523.
- [4] Jas Brooks, Alex Mazursky, Janice Hixon, and Pedro Lopes. 2024. Augmented Breathing via Thermal Feedback in the Nose. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–11.
- [5] Antonio Cataldo, Elisa Raffaella Ferrè, Giuseppe Di Pellegrino, and Patrick Haggard. 2016. Thermal referral: evidence for a thermoceptive uniformity illusion without touch. *Scientific reports* 6, 1 (2016), 1–10. doi:10.1038/nrn755
- [6] Zikun Chen, Roshan Lalintha Peiris, and Kouta Minamizawa. 2017. A thermal pattern design for providing dynamic thermal feedback on the face with head mounted displays. In *Proceedings of the Eleventh International Conference on Tangible, Embedded, and Embodied Interaction*. 381–388.
- [7] Ian Darian-Smith and Kenneth O. Johnson. 1977. Thermal sensibility and thermoreceptors. *Journal of Investigative Dermatology* 69, 1 (1977), 146–153. doi:10.1111/1523-1747.ep12497936
- [8] Ian Darian-Smith and Kenneth O. Johnson. 1977. THERMAL SENSIBILITY AND THERMORECEPTORS. *Journal of Investigative Dermatology* 69, 1 (1977), 146–153. doi:10.1111/1523-1747.ep12497936
- [9] Peter James Dyck, I. Zimmerman, DA Gillen, D. Johnson, JL Karnes, and PC O’Brien. 1993. Cool, warm, and heat-pain detection thresholds: testing methods and inferences about anatomic distribution of receptors. *Neurology* 43, 8 (1993), 1500–1500. doi:10.1212/WNL.43.8.1500
- [10] Abdallah El Ali, Xingyu Yang, Swamy Ananthanarayan, Thomas Rögglä, Jack Jansen, Jess Hartcher-O’Brien, Kaspar Jansen, and Pablo Cesar. 2020. ThermalWear: Exploring Wearable On-Chest Thermal Displays to Augment Voice Messages with Affect. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI ’20). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3313831.3376682
- [11] Abdallah El Ali, Xingyu Yang, Swamy Ananthanarayan, Thomas Rögglä, Jack Jansen, Jess Hartcher-O’Brien, Kaspar Jansen, and Pablo Cesar. 2020. Thermal-wear: Exploring wearable on-chest thermal displays to augment voice messages with affect. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [12] Talia Sofia Ezer, Jonathan Giron, Hadas Erel, and Oren Zuckerman. 2024. Somaesthetic Meditation Wearable: Exploring the Effect of Targeted Warmth Technology on Meditators’ Experiences. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–14.
- [13] Davide Filingeri. 2016. Neurophysiology of skin thermal sensations. (2016). doi:10.1111/1523-1747.ep12497936
- [14] Massimiliano Gabardi, Daniele Leonardi, Massimiliano Solazzi, and Antonio Frisoli. 2018. Development of a miniaturized thermal module designed for integration in a wearable haptic device. In *2018 IEEE Haptics Symposium (HAPTICS)*. IEEE, 100–105.
- [15] Daniel Gongora, Roshan Lalintha Peiris, and Kouta Minamizawa. 2017. Towards intermanual apparent motion of thermal pulses. In *Adjunct Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology*. 143–145.
- [16] Barry G. Green. 1977. Localization of thermal sensation: An illusion and synthetic heat. *Perception & Psychophysics* 22, 4 (1977), 331–337. doi:10.3758/BF03199698
- [17] Barry G. Green. 1978. Referred thermal sensations: warmth versus cold. *Sensory processes* 2, 3 (1978), 220–230.
- [18] Sebastian Günther, Alexandra Skogseide, Robin Buhlmann, and Max Mühlhäuser. 2024. Assessing the Influence of Visual Cues in Virtual Reality on the Spatial Perception of Physical Thermal Stimuli. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–12.
- [19] Teng Han, Sirui Wang, Sijia Wang, Xiangmin Fan, Jie Liu, Feng Tian, and Mingming Fan. 2020. Mouillé: Exploring wetness illusion on fingertips to enhance immersive experience in vr. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–10.
- [20] Jeanne Hecquard, Justine Saint-Aubert, Ferran Argelaguet, Claudio Pacchierotti, Anatole Lécuyer, and Marc JM Macé. 2024. Warm regards: Influence of thermal haptic feedback during social interactions in VR. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 1216–1225.
- [21] Hsin-Ni Ho, Junji Watanabe, Hideyuki Ando, and Makio Kashino. 2011. Mechanisms underlying referral of thermal sensations to sites of tactile stimulation. *Journal of Neuroscience* 31, 1 (2011), 208–213. doi:10.1523/JNEUROSCI.2640-10.2011
- [22] Ali Israr and Ivan Poupyrev. 2011. Tactile brush: drawing on skin with a tactile grid display. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2019–2028.
- [23] K.P. Ivanov. 1999. Subject of temperature control and the main function of thermoregulation of an organism. *Journal of Thermal Biology* 24, 5 (1999), 415–421. doi:10.1016/S0306-4565(99)00060-1

- [24] Takuya Jodai, Masahiko Terao, Lynette A Jones, and Hsin-Ni Ho. 2023. Determination of the thermal-tactile simultaneity window for multisensory cutaneous displays. In *2023 IEEE World Haptics Conference (WHC)*. IEEE, 230–236.
- [25] Lynette A Jones and Hsin-Ni Ho. 2008. Warm or cool, large or small? The challenge of thermal displays. *IEEE Transactions on Haptics* 1, 1 (2008), 53–70. doi:10.1109/TOH.2008.2
- [26] Lynette A Jones and Hsin-Ni Ho. 2025. Tactile–Thermal Interactions: Cooperation and Competition. *IEEE Transactions on Haptics* (2025).
- [27] Seongjun Kang, Gwangbin Kim, Seokhyun Hwang, Jeongju Park, Ahmed Ibrahim Ahmed Mohamed Elsharkawy, and SeungJun Kim. 2024. Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–15.
- [28] Jae-Hwan Kim, Abraham Vázquez-Guardado, Haiwen Luan, Jin-Tae Kim, Da Som Yang, Hao-hui Zhang, Jan-Kai Chang, Seonggwang Yoo, Chanho Park, Yuan-ting Wei, et al. 2024. A wirelessly programmable, skin-integrated thermo-haptic stimulator system for virtual reality. *Proceedings of the National Academy of Sciences* 121, 22 (2024), e2404007121.
- [29] Seongho Kim, Taeyeon Kim, Choong Sun Kim, Hyeongdo Choi, Yong Jun Kim, Gyu Soup Lee, Ockkyun Oh, and Byung Jin Cho. 2020. Two-dimensional thermal haptic module based on a flexible thermoelectric device. *Soft Robotics* 7, 6 (2020), 736–742.
- [30] Jacob H Kirman. 1974. Tactile apparent movement: The effects of interstimulus onset interval and stimulus duration. *Perception & Psychophysics* 15, 1 (1974), 1–6.
- [31] Sven Kratz and Tony Dunnigan. 2017. Thermotouch: a new scalable hardware design for thermal displays. In *Proceedings of the 2017 ACM International Conference on Interactive Surfaces and Spaces*. 132–141.
- [32] Inès Lacôte, David Gueorguiev, Claudio Pacchierotti, Marie Babel, and Maud Marchal. 2022. Speed discrimination in the apparent haptic motion illusion. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*. Springer, 48–56.
- [33] Zhuying Li, Yan Wang, Wei Wang, Weikang Chen, Ti Hoang, Stefan Greuter, and Florian Floyd Mueller. 2019. HeatCraft: Designing playful experiences with ingestible sensors via localized thermal stimuli. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [34] Yuhu Liu, Satoshi Nishikawa, Young ah Seong, Ryuma Niiyama, and Yasuo Kuniyoshi. 2021. ThermoCaress: A Wearable Haptic Device with Illusory Moving Thermal Stimulation. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (, Yokohama, Japan,) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 214, 12 pages. doi:10.1145/3411764.3445777
- [35] Maohui Luo, Zhe Wang, Hui Zhang, Edward Arens, Davide Filingeri, Ling Jin, Ali Ghahramani, Wenhua Chen, Yingdong He, and Binghui Si. 2020. High-density thermal sensitivity maps of the human body. *Building and environment* 167 (2020), 106435.
- [36] Tim Moesgen, Hsin-Ni Ho, and Yu Xiao. 2024. Apparent Thermal Motion on the Forearm. In *Eurohaptics 2024 (Lecture Notes in Computer Science)*. Springer. | openaire: EC/HE/101070533/EU/EMIL; Eurohaptics Conference ; Conference date: 30-06-2024 Through 03-07-2024.
- [37] Tim Moesgen, Hsin-Ni Ho, and Yu Xiao. 2024. Apparent thermal motion on the forearm. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*. Springer, 56–68.
- [38] Jaeyoung Park, Jaeha Kim, Yonghwan Oh, and Hong Z Tan. 2016. Rendering moving tactile stroke on the palm using a sparse 2d array. In *Haptics: Perception, Devices, Control, and Applications: 10th International Conference, EuroHaptics 2016, London, UK, July 4-7, 2016, Proceedings, Part I* 10. Springer, 47–56.
- [39] Roshan Lalintha Peiris, Wei Peng, Zikun Chen, Liwei Chan, and Kouta Minamizawa. 2017. ThermoVR: Exploring Integrated Thermal Haptic Feedback with Head Mounted Displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 5452–5456. doi:10.1145/3025453.3025824
- [40] Luka Peters, Gokhan Serhat, and Yasemin Vardar. 2023. ThermoSurf: Thermal display technology for dynamic and multi-finger interactions. *Ieee Access* 11 (2023), 12004–12014.
- [41] Oliver S Schneider, Ali Israr, and Karon E MacLean. 2015. Tactile animation by direct manipulation of grid displays. In *Proceedings of the 28th annual ACM symposium on user interface software & technology*. 21–30.
- [42] Yosef Y Shani and Simon Lineykin. 2023. Thermal Cues Composed of Sequences of Pulses for Transferring Data via a Haptic Thermal Interface. *Bioengineering* 10, 10 (2023), 1156.
- [43] Carl E Sherrick and Ronald Rogers. 1966. Apparent haptic movement. *Perception & Psychophysics* 1, 3 (1966), 175–180.
- [44] Yatharth Singhal, Daniel Honrales, Hsin-Ni Ho, and Jin Ryong Kim. 2024. Wetness Illusion in Mid-Air. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 846–854.
- [45] Yatharth Singhal, Daniel Honrales, Haokun Wang, and Jin Ryong Kim. 2024. Thermal in motion: Designing thermal flow illusions with tactile and thermal interaction. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–13.
- [46] Yatharth Singhal, Haokun Wang, Hyunjae Gil, and Jin Ryong Kim. 2021. Mid-Air Thermo-Tactile Feedback Using Ultrasound Haptic Display. In *Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology* (Osaka, Japan) (VRST '21). Association for Computing Machinery, New York, NY, USA, Article 28, 11 pages. doi:10.1145/3489849.3489889
- [47] Hyunki Son, Haokun Wang, Yatharth Singhal, and Jin Ryong Kim. 2023. Upper Body Thermal Referral and Tactile Masking for Localized Feedback. *IEEE Transactions on Visualization and Computer Graphics* 29, 5 (2023), 2211–2219. doi:10.1109/TVCG.2023.3247068
- [48] Zhongda Sun, Zixuan Zhang, and Chengkuo Lee. 2023. A skin-like multimodal haptic interface. *Nature Electronics* 6, 12 (2023), 941–942.
- [49] Toki Takeda, Arinobu Nijima, Takafumi Mukouchi, and Takashi Satou. 2020. Creating illusion of wind blowing with air vortex-induced apparent tactile motion. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–7.
- [50] Chan Lek Tan and Zachary A. Knight. 2018. Regulation of Body Temperature by the Nervous System. *Neuron* 98, 1 (2018), 31–48. doi:10.1016/j.neuron.2018.02.022
- [51] Muhammad Umair, Corina Sas, Niaz Chahabianloo, and Cem Ersoy. 2021. Exploring personalized vibrotactile and thermal patterns for affect regulation. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference*. 891–906.
- [52] Marika Vellei, Richard de Dear, Christian Inard, and Ollie Jay. 2021. Dynamic thermal perception: A review and agenda for future experimental research. *Building and Environment* 205 (2021), 108269. doi:10.1016/j.buildenv.2021.108269
- [53] Marika Vellei, Richard De Dear, Christian Inard, and Ollie Jay. 2021. Dynamic thermal perception: A review and agenda for future experimental research. *Building and Environment* 205 (2021), 108269. doi:10.1016/j.buildenv.2021.108269
- [54] Steeven Villa, Kenji Ishihara, Moritz Ziarko, Sebastian Günther, and Florian Müller. 2024. Touch It Like It's Hot: A Thermal Feedback Enabled Encountered-type Haptic Display for Virtual Reality. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 700–709.
- [55] Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. 2024. Fiery hands: Designing thermal glove through thermal and tactile integration for virtual object manipulation. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*. 1–15.
- [56] Haokun Wang, Yatharth Singhal, Hyunjae Gil, and Jin Ryong Kim. 2024. Thermal Masking: When the Illusion Takes Over the Real. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–16.
- [57] Haokun Wang, Yatharth Singhal, and Jin Ryong Kim. 2023. Fabric Thermal Display using Ultrasonic Waves. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 336–345.
- [58] Haokun Wang, Yatharth Singhal, and Jin Ryong Kim. 2024. Let It Snow: Designing Snowfall Experience in VR. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 8, 2 (2024), 1–24.
- [59] Ryo Watanabe, Ryuta Okazaki, and Hiroyuki Kajimoto. 2014. Mutual referral of thermal sensation between two thermal-tactile stimuli. In *2014 IEEE haptics symposium (haptics)*. IEEE, 299–302. doi:10.1109/HAPTICS.2014.6775471
- [60] Connor Wilson, Daniel J Rea, and Scott Bateman. 2022. Temperature Illusions in Mixed Reality using Color and Dynamic Graphics. In *Graphics Interface 2023*.
- [61] Graham Wilson, Stephen Brewster, Martin Halvey, and Stephen Hughes. 2012. Thermal icons: evaluating structured thermal feedback for mobile interaction. In *Proceedings of the 14th international conference on Human-computer interaction with mobile devices and services*. 309–312.
- [62] Graham Wilson and Stephen A Brewster. 2017. Multi-moji: Combining thermal, vibrotactile & visual stimuli to expand the affective range of feedback. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 1743–1755.
- [63] Weicheng Wu and Heather Culbertson. 2019. Wearable haptic pneumatic device for creating the illusion of lateral motion on the arm. In *2019 IEEE World Haptics Conference (WHC)*. IEEE, 193–198.
- [64] Rui Xiao and X.Z. Shawn Xu. 2021. Temperature Sensation: From Molecular Thermosensors to Neural Circuits and Coding Principles. *Annual Review of Physiology* 83, 1 (2021), 205–230. doi:10.1146/annurev-physiol-031220-095215 PMID: 33085927.
- [65] Hyung Li Yi, Hojeong Lee, and Sang Ho Yoon. 2024. ThermicVib: Enabling Dynamic Thermal Sensation with Multimodal Haptic Glove for Thermal-Responsive Interaction. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 614–623.